CENTRAL CALIFORNIA
EMERGENCY MEDICAL SERVICES

A Division of the Fresno County Department of Public Health

Manual	Policy
Emergency Medical Services Administrative Policies and Procedures	Number 530.33
Subject Paramedic Treatment Protocols **PEDIATRIC ARREST- MEDICAL**	
References Title 22, Division 9, Chapter 3.3 of the California Code of Regulations	Effective Fresno County: 01/15/82 Kings County: 04/10/89 Madera County: 06/15/85 Tulare County: 04/19/05

STANDING ORDERS

Assess need for CPR	Palpate femoral or brachial pulse for no longer than 10 seconds. If in doubt, begin CPR.
Compressions	Perform high quality manual chest compressions (see special considerations)
Airway	Secure with BLS airway and suction as needed.
Ventilation and Oxygen	Ventilate with high flow O2 while patient in arrest. Ventilation and compression ratio 15:2 with two providers, 30:2 with one provider. Ventilate to achieve chest rise.
Assess Rhythm	- Apply cardiac monitor and defibrillator pads as soon as possible. - Use pediatric pads in age ≤8 years. Defibrillator pads must not overlap. - Treat according to rhythm. Defibrillate immediately if VF/VT.
Waveform capnography	Monitor with continuous waveform capnography to monitor effectiveness of compressions and ventilations.
IV/IO Access	LR TKO with Volutrol and Pediatric tubing.

PEA/Asystole		Ventricular Fibrillation/Ventricular Tachycardia	
Epinephrine	0.01 mg/kg (0.1ml/kg) of 1:10,000 IV/IO	Defibrillate	As soon as possible and every 2 minutes while in VF/VT 1st attempt: 2 J/kg Subsequent attempts: 4 J/kg
Fluid Bolus	20 ml/kg NS or LR for suspected hypovolemia.	Epinephrine	0.01 mg/kg (0.1 ml/kg) of 1:10,000 IV/IO. After defibrillation x 2. Repeat every 3-5 minutes.
		Amiodarone	For refractory VF/VT. After defibrillation x3: 5 mg/kg IV/IO. Max single dose 300 mg. May repeat x2. Total max dose 450 mg.

STANDING ORDERS - CONTINUED ON NEXT PAGE

Approved By		Revision
EMS Director	**Daniel J. Lynch** (Signature on File at EMS Agency)	**09/01/2025**
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STANDING ORDERS (CONTINUED)	
Transport	Per EMS policy #547. Initiate transport after initial attempts at stabilization (airway, defibrillation, IV/IO access, one round of ACLS drugs).
Post-Arrest Care/Return of Spontaneous Circulation	
Assess blood pressure	Recheck every 3-5 minutes at minimum.
Oxygen	Titrate to maintain O2 sats 94-99%
Waveform capnography	Monitor continuously. Target EtCO2 is 35-45 mmHg. Sudden decrease in EtCO2 is suggestive of rearrest.
IV fluid bolus	20 ml/kg if hypotension for age or signs of shock (capillary refill > 2 seconds, weak/thready pulses). May repeat x 1.
Check blood glucose	If < 60 mg/dL: - Dextrose 10% 5ml/kg IV/IO. Max dose 250 ml - Glucagon IM (if no IV/IO access): 0.5 mg (0.5 ml) if < 25 kg 1 mg (1 ml) if > 25 kg
If patient rearrests	Immediately resume compressions and treat according to rhythm above.

SPECIAL CONSIDERATIONS AND PRIORITIES

1. This protocol applies to patients ≤ 14 years of age. Neonates born in the field should be resuscitated according to EMS Protocol # 530.31. Patients > 14 years of age should be treated according to EMS Protocol # 530.03.

2. Refer to Length Based Pediatric Tape to estimate weight and refer to drug tables below for specific doses. If patient exceeds the length of the tape, refer to adult doses.

3. High performance CPR includes the following elements:
   - Compress at 110 compressions per minute. Use of metronome is recommended
   - Ensure adequate depth: 1.5 inches in infants, 2 inches in children
   - Allow adequate recoil between compressions. Rotate compressors every 2 minutes to prevent fatigue.
   - Minimize interruptions. Goal >90% compression fraction. Pre-charge monitor prior to pulse checks and do not pause CPR to charge or perform advanced interventions.
   - Do not over ventilate. Provide only enough ventilation to achieve chest rise.

4. Most pediatric arrests are due to respiratory failure. Focus on providing quality ventilations and oxygenation, ensuring chest rise with each breath.

5. Hyperoxia may lead to worse outcomes and neurologic injury in patients with ROSC. High flow O2 is indicated for patients in arrest. However, in the post-arrest phase of care, oxygen should be titrated to O2 target O2 saturations 94-99%. If possible, wean O2 if SpO2 is 100%.

6. Minimum systolic blood pressure for patients > 1 year can be calculated with the following formula:
   70 mmHg + (2 x age in years)

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GRAY (3-5 kg) **REFER TO PROTOCOL #530.01 NEONATAL RESUSCITATION FOR INFANTS BORN IN THE FIELD**			
Approximate age	<3 months	Respiratory Rate	30-60
Heart rate	100-180	Minimum SBP	60
EPINEPHRINE IV/IO			
Concentration 0.1mg/ml (1:10,000)	**Calculated Dose** 0.04 mg	**Volume** 0.4 ml	
AMIODARONE (IF VF/VT)			
NOT INDICATED			
DEFIBRILLATE (IF VT/VT)			
1st Attempt	**8 joules**		
Subsequent Attempts	**16 joules**		
FLUID BOLUS			
NS or LR	**80 ml**		
HYPOGLYCEMIA MANAGEMENT (POST-ARREST)			
Dextrose 10% IV/IO	**20 ml slow IV push**		
Glucagon IM	**0.5 mg**	**0.5 ml**	

PINK (6-7 kg)			
Approximate age	4 months	Respiratory Rate	30-60
Heart rate	100-180	Minimum SBP	70
EPINEPHRINE IV/IO			
Concentration 0.1mg/ml (1:10,000)	**Calculated Dose** 0.06 mg	**Volume** 0.6 ml	
AMIODARONE (IF VF/VT)			
50 mg/ml	**30 mg**	**0.6 ml**	
DEFIBRILLATE (IF VT/VT)			
1st Attempt	**10 J**		
Subsequent Attempts	**20 J**		
FLUID BOLUS			
NS or LR	**120 ml**		
HYPOGLYCEMIA MANAGEMENT (POST-ARREST)			
Dextrose 10%	**30 ml slow IV push**		
Glucagon IM	**0.5 mg**	**0.5 ml**	

RED (8-9 kg)			
Approximate age	8 months	Respiratory Rate	30-60
Heart rate	100-180	Minimum SBP	70
EPINEPHRINE IV/IO			
Concentration 0.1mg/ml (1:10,000)	**Calculated Dose** 0.08 mg	**Volume** 0.8 ml	
AMIODARONE (IF VF/VT)			
50 mg/ml	**40 mg**	**0.8 ml**	
DEFIBRILLATE (IF VT/VT)			
1st Attempt	**15 J**		
Subsequent Attempts	**30 J**		
FLUID BOLUS			
NS or LR	**160 ml**		
HYPOGLYCEMIA MANAGEMENT (POST-ARREST)			
Dextrose 10%	**40 ml slow IV push**		
Glucagon IM	**0.5 mg**	**0.5 ml**	

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PURPLE (10-11 kg)			
Approximate age Heart rate	12 months 90-170	Respiratory Rate Minimum SBP	24-40 72
EPINEPHRINE IV/IO			
Concentration **0.1mg/ml (1:10,000)**	**Calculated Dose** **0.1 mg**	**Volume** **1 ml**	
AMIODARONE (IF VF/VT)			
50 mg/ml	**50 mg**	**1 ml**	
DEFIBRILLATE (IF VT/VT)			
1st Attempt	**20 J**		
Subsequent Attempts	**40 J**		
FLUID BOLUS			
NS or LR	**200 ml**		
HYPOGLYCEMIA MANAGEMENT (POST-ARREST)			
Dextrose 10%	**50 ml slow IV push**		
Glucagon IM	**1 mg**	**1 ml**	

YELLOW (12-14 kg)			
Approximate age Heart rate	2 years 90-160	Respiratory Rate Minimum SBP	24-40 74
EPINEPHRINE IV/IO			
Concentration **0.1mg/ml (1:10,000)**	**Calculated Dose** **0.13 mg**	**Volume** **1.3 ml**	
AMIODARONE (IF VF/VT)			
50 mg/ml	**65 mg**	**1.3 ml**	
DEFIBRILLATE (IF VT/VT)			
1st Attempt	**30 J**		
Subsequent Attempts	**50 J**		
FLUID BOLUS			
NS or LR	**260 ml**		
HYPOGLYCEMIA MANAGEMENT (POST-ARREST)			
Dextrose 10%	**65 ml slow IV push**		
Glucagon IM	**0.5 mg**	**0.5 ml**	

WHITE (15-18 kg)			
Approximate age Heart rate	4 years 80-140	Respiratory Rate Minimum SBP	22-34 78
EPINEPHRINE IV/IO			
Concentration **0.1mg/ml (1:10,000)**	**Calculated Dose** **0.16 mg**	**Volume** **1.6 ml**	
AMIODARONE (IF VF/VT)			
50 mg/ml	**80 mg**	**1.6 ml**	
DEFIBRILLATE (IF VT/VT)			
1st Attempt	**30 J**		
Subsequent Attempts	**70 J**		
FLUID BOLUS			
NS or LR	**320 ml**		
HYPOGLYCEMIA MANAGEMENT (POST-ARREST)			
Dextrose 10%	**80 ml slow IV push**		
Glucagon	**0.5 mg**	**0.5 ml**	

BLUE (19-23 kg)		
Approximate age 6 years Heart rate 70-130	Respiratory Rate 20-30 Minimum SBP 82	
EPINEPHRINE IV/IO		
Concentration **0.1mg/ml (1:10,000)**	**Calculated Dose** **0.2 mg**	**Volume** **2.0 ml**
AMIODARONE (IF VF/VT)		
50 mg/ml	**100 mg**	**2.0 ml**
DEFIBRILLATE (IF VT/VT)		
1st Attempt	**40 J**	
Subsequent Attempts	**80 J**	
FLUID BOLUS		
NS or LR	**400 ml**	
HYPOGLYCEMIA MANAGEMENT (POST-ARREST)		
Dextrose 10%	**100 ml**	
Glucagon IM	**0.5 mg**	**0.5 ml**

ORANGE (24-29 kg)		
Approximate age 8 years Heart rate 70-130	Respiratory Rate 18-30 Minimum SBP 86	
EPINEPHRINE IV/IO		
Concentration **0.1mg/ml (1:10,000)**	**Calculated Dose** **0.27 mg**	**Volume** **2.7 ml**
AMIODARONE (IF VF/VT)		
50 mg/ml	**135 mg**	**2.7 ml**
DEFIBRILLATE (IF VT/VT)		
1st Attempt	**55 J**	
Subsequent Attempts	**110 J**	
FLUID BOLUS		
NS or LR	**500 ml**	
HYPOGLYCEMIA MANAGEMENT (POST-ARREST)		
Dextrose 10%	**135 ml**	
Glucagon IM	**1 mg**	**1 ml**

GREEN (30-36 kg)		
Approximate age 10 years Heart rate 60-110	Respiratory Rate 18-30 Minimum SBP 90	
EPINEPHRINE IV/IO		
Concentration **0.1mg/ml (1:10,000)**	**Calculated Dose** **0.32 mg**	**Volume** **3.2 ml**
AMIODARONE (IF VF/VT)		
50 mg/ml	**160 mg**	**3.2 ml**
DEFIBRILLATE (IF VT/VT)		
1st Attempt	**70 joules**	
Subsequent Attempts	**130 joules**	
FLUID BOLUS		
NS or LR	**640 ml**	
HYPOGLYCEMIA MANAGEMENT (POST-ARREST)		
Dextrose 10%	**160 ml**	
Glucagon IM	**1 mg**	**1 ml**